SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Project Title:

Bookswagon – Online Book Store Order Management Database System

1. BASIC SQL QUERIES

1.1 SELECT OUERY

The SELECT statement is used to retrieve data from table.

Task 1.1.1 – Display All Books

```
SQL> SELECT *
  2  FROM Book;

BOOK_ID TITLE                          AUTHOR              CATEGORY_ID       PRICE STOCK
------- ------------------------------ ------------------- ----------- ----------- -----
    101 The Alchemist                  Paulo Coelho                  1      299.00    50
    102 Atomic Habits                  James Clear                   2      499.00    35
    103 Database System Concepts       Silberschatz                  3      850.00    25
    104 Panchatantra Stories           Vishnu Sharma                 4      199.00    40
    106 The Kite Runner                Khaled Hosseini               1      350.00    30
    107 Think and Grow Rich            Napoleon Hill                 2      275.00    45
    108 Let Us C                       Yashavant Kanetkar            3      450.00    20
    109 Wings of Fire                  APJ Abdul Kalam               2      325.00    55
    110 Harry Potter Stone             J.K. Rowling                  1      599.00    30
    112 Fundamentals of DBMS           Elmasri Navathe               3      950.00    28
    113 The Jungle Book                Rudyard Kipling               4      225.00    22
    114 Ponniyin Selvan                Kalki                         1      699.00    40

12 rows selected.
```

Task 1.1.2 – Display Selected Book Details

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book;

BOOK_ID TITLE                              PRICE
------- ------------------------------ ----------
    101 The Alchemist                      299.00
    102 Atomic Habits                      499.00
    103 Database System Concepts           850.00
    104 Panchatantra Stories               199.00
    106 The Kite Runner                    350.00
    107 Think and Grow Rich                275.00
    108 Let Us C                           450.00
    109 Wings of Fire                      325.00
    110 Harry Potter Stone                 599.00
    112 Fundamentals of DBMS               950.00
    113 The Jungle Book                    225.00
    114 Ponniyin Selvan                    699.00

12 rows selected.
```

Task 1.1.3 – Find Books with Price Less Than ₹ 1000

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book
  3  WHERE Price < 1000;

BOOK_ID TITLE                              PRICE
------- ------------------------------ ----------
    101 The Alchemist                      299.00
    102 Atomic Habits                      499.00
    103 Database System Concepts           850.00
    104 Panchatantra Stories               199.00
    106 The Kite Runner                    350.00
    107 Think and Grow Rich                275.00
    108 Let Us C                           450.00
    109 Wings of Fire                      325.00
    110 Harry Potter Stone                 599.00
    112 Fundamentals of DBMS               950.00
    113 The Jungle Book                    225.00
    114 Ponniyin Selvan                    699.00

12 rows selected.
```

Task 1.1.4 – Find Books with Price Greater Than ₹ 100

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book
  3  WHERE Price > 100;

BOOK_ID TITLE                                PRICE
------- ------------------------------ -----------
    101 The Alchemist                       299.00
    102 Atomic Habits                       499.00
    103 Database System Concepts            850.00
    104 Panchatantra Stories                199.00
    106 The Kite Runner                     350.00
    107 Think and Grow Rich                 275.00
    108 Let Us C                            450.00
    109 Wings of Fire                       325.00
    110 Harry Potter Stone                  599.00
    112 Fundamentals of DBMS                950.00
    113 The Jungle Book                     225.00
    114 Ponniyin Selvan                     699.00

12 rows selected.
```

Task 1.1.5 – Books from Lowest to Highest Price

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book
  3  ORDER BY Price ASC;

BOOK_ID TITLE                                PRICE
------- ------------------------------ -----------
    104 Panchatantra Stories                199.00
    113 The Jungle Book                     225.00
    107 Think and Grow Rich                 275.00
    101 The Alchemist                       299.00
    109 Wings of Fire                       325.00
    106 The Kite Runner                     350.00
    108 Let Us C                            450.00
    102 Atomic Habits                       499.00
    110 Harry Potter Stone                  599.00
    114 Ponniyin Selvan                     699.00
    103 Database System Concepts            850.00
    112 Fundamentals of DBMS                950.00

12 rows selected.
```

Task 1.1.6 – Books from Highest to Lowest Price

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book
  3  ORDER BY Price DESC;

BOOK_ID TITLE                              PRICE
------- ------------------------------ ----------
    112 Fundamentals of DBMS               950.00
    103 Database System Concepts           850.00
    114 Ponniyin Selvan                    699.00
    110 Harry Potter Stone                 599.00
    102 Atomic Habits                      499.00
    108 Let Us C                           450.00
    106 The Kite Runner                    350.00
    109 Wings of Fire                      325.00
    101 The Alchemist                      299.00
    107 Think and Grow Rich                275.00
    113 The Jungle Book                    225.00
    104 Panchatantra Stories               199.00

12 rows selected.
```

Task 1.1.7 – Display Unique Book Categories

```
SQL> SELECT DISTINCT
  2      c.Category_ID,
  3      c.Category_Name
  4  FROM Book b
  5  JOIN Category c
  6      ON b.Category_ID = c.Category_ID
  7  ORDER BY c.Category_ID;

CATEGORY_ID CATEGORY_NAME
----------- --------------------
          1 Fiction
          2 Self-Help
          3 Academic
          4 Children

4 rows selected.
```

2. SEARCH BOOKS

Task 2.1 – Search Books Based on Price

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book
  3  WHERE Price BETWEEN 100 AND 400
  4  ORDER BY Price ASC;

BOOK_ID TITLE                                PRICE
------- ---------------------------- -----------
    104 Panchatantra Stories              199.00
    113 The Jungle Book                   225.00
    107 Think and Grow Rich               275.00
    101 The Alchemist                     299.00
    109 Wings of Fire                     325.00
    106 The Kite Runner                   350.00

6 rows selected.
```

Task 2.2– Search Books Based on Category

```
SQL> SELECT
  2      b.Book_ID,
  3      b.Title,
  4      c.Category_Name,
  5      b.Price
  6  FROM Book b
  7  JOIN Category c
  8      ON b.Category_ID = c.Category_ID
  9  WHERE c.Category_ID = 1;

BOOK_ID TITLE                        CATEGORY_NAME             PRICE
------- ---------------------------- ------------------ -----------
    101 The Alchemist                Fiction                299.00
    106 The Kite Runner              Fiction                350.00
    110 Harry Potter Stone           Fiction                599.00
    114 Ponniyin Selvan              Fiction                699.00

4 rows selected.
```

3. RETRIEVE CUSTOMER INFORMATION

Task 3.1 – Display All Customers

```
SQL> SELECT *
  2  FROM Customer;

CUSTOMER_ID CUSTOMER_NAME                 EMAIL                          PHONE_NO ADDRESS
----------- ----------------------------- ------------------------------ ---------- --------------
          1 Nivethitha                    nivethitha@gmail.com           9876543210 Chennai
          2 Hema                          hema@gmail.com                 9876501234 Chennai
          3 Ravi                          ravi@gmail.com                 9087654321 Chengalpattu
          4 Karthik                       karthik@gmail.com              9876543211 Coimbatore
          5 Priya                         priya@gmail.com                8765432109 Salem
          6 Divya                         divya@gmail.com                7654321098 Trichy
          7 Arun                          arun@gmail.com                 6543210987 Vellore
          8 Meena                         meena@gmail.com                9432109876 Tirunelveli
          9 Nisha                         nisha@gmail.com                9210987654 Madurai

9 rows selected.
```

Task 3.2– Search for a Particular Customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 1;

CUSTOMER_ID CUSTOMER_NAME                 EMAIL                          PHONE_NO ADDRESS
----------- ----------------------------- ------------------------------ ---------- --------------
          1 Nivethitha                    nivethitha@gmail.com           9876543210 Chennai

1 row selected.
```

4. APPLY FILTERING CONDITIONS

Multiple conditions can be combined using AND, OR, and NOT.

Task 4.1– Apply AND Condition

```
SQL> SELECT Book_ID, Title, Price, Stock
  2  FROM Book
  3  WHERE Price < 500
  4  AND Stock > 30;

BOOK_ID TITLE                               PRICE STOCK
------- ------------------------------ ---------- -----
    101 The Alchemist                      299.00    50
    102 Atomic Habits                      499.00    35
    104 Panchatantra Stories               199.00    40
    107 Think and Grow Rich                275.00    45
    109 Wings of Fire                      325.00    55

5 rows selected.
```

Task 4.2 – Apply OR Condition

```
SQL> SELECT
  2      b.Book_ID,
  3      b.Title,
  4      c.Category_Name,
  5      b.Price
  6  FROM Book b
  7  JOIN Category c
  8      ON b.Category_ID = c.Category_ID
  9  WHERE b.Category_ID = 1
 10      OR b.Category_ID = 4;

BOOK_ID TITLE                          CATEGORY_NAME              PRICE
------- ---------------------------- ------------------ -----------
    101 The Alchemist                Fiction                  299.00
    106 The Kite Runner              Fiction                  350.00
    110 Harry Potter Stone           Fiction                  599.00
    114 Ponniyin Selvan              Fiction                  699.00
    104 Panchatantra Stories         Children                 199.00
    113 The Jungle Book              Children                 225.00

6 rows selected.
```

Task 4.3 – Apply IN Operator

```
SQL> SELECT
  2      b.Book_ID,
  3      b.Title,
  4      c.Category_Name,
  5      b.Price
  6  FROM Book b
  7  JOIN Category c
  8      ON b.Category_ID = c.Category_ID
  9  WHERE b.Category_ID IN (2, 3);

BOOK_ID TITLE                          CATEGORY_NAME              PRICE
------- ---------------------------- ------------------ -----------
    102 Atomic Habits                Self-Help                499.00
    107 Think and Grow Rich          Self-Help                275.00
    109 Wings of Fire                Self-Help                325.00
    103 Database System Concepts     Academic                 850.00
    108 Let Us C                     Academic                 450.00
    112 Fundamentals of DBMS         Academic                 950.00

6 rows selected.
```

Task 4.4– Apply BETWEEN Operator

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book
  3  WHERE Price BETWEEN 100 AND 400
  4  ORDER BY Price ASC;

BOOK_ID TITLE                          PRICE
------- ---------------------------- -----------
    104 Panchatantra Stories                199.00
    113 The Jungle Book                     225.00
    107 Think and Grow Rich                 275.00
    101 The Alchemist                       299.00
    109 Wings of Fire                       325.00
    106 The Kite Runner                     350.00

6 rows selected.
```

5. BASIC BUSINESS REPORTS

Task 5.1 – Book Price Report

```
SQL> SELECT Book_ID, Title, Price
  2  FROM Book
  3  ORDER BY Price DESC;

BOOK_ID TITLE                          PRICE
------- ---------------------------- -----------
    112 Fundamentals of DBMS              950.00
    103 Database System Concepts          850.00
    114 Ponniyin Selvan                   699.00
    110 Harry Potter Stone                599.00
    102 Atomic Habits                     499.00
    108 Let Us C                          450.00
    106 The Kite Runner                   350.00
    109 Wings of Fire                     325.00
    101 The Alchemist                     299.00
    107 Think and Grow Rich               275.00
    113 The Jungle Book                   225.00
    104 Panchatantra Stories              199.00

12 rows selected.
```

Task 5.2 – Customer Order Report

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount,
  7      o.Order_Status
  8  FROM Customer c
  9  JOIN Orders o
 10      ON c.Customer_ID = o.Customer_ID
 11  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID CUSTOMER_NAME                ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- ---------------------------- -------- --------- ------------ ---------------
          1 Nivethitha                       1001 12-OCT-26          873 Delivered
          2 Hema                             1002 12-OCT-26         1149 Shipped
          3 Ravi                             1003 13-OCT-26          850 Processing
          4 Karthik                          1004 13-OCT-26          947 Delivered
          5 Priya                            1005 14-OCT-26         1049 Shipped
          6 Divya                            1006 14-OCT-26          950 Processing
          7 Arun                             1007 15-OCT-26         1149 Delivered
          8 Meena                            1008 15-OCT-26         1297 Processing
          9 Nisha                            1009 16-OCT-26          550 Processing

9 rows selected.
```

Task 5.3 – Book Order Report

```
SQL> SELECT
  2      b.Book_ID,
  3      b.Title,
  4      o.Order_ID,
  5      oi.Quantity,
  6      oi.Price,
  7      oi.Total_Amount
  8  FROM Book b
  9  JOIN Order_Item oi
 10      ON b.Book_ID = oi.Book_ID
 11  JOIN Orders o
 12      ON oi.Order_ID = o.Order_ID
 13  ORDER BY b.Book_ID, o.Order_ID;

BOOK_ID TITLE                          ORDER_ID QUANTITY       PRICE TOTAL_AMOUNT
------- ------------------------------ -------- -------- ----------- ------------
    101 The Alchemist                      1001        2      299.00          598
    101 The Alchemist                      1008        1      299.00          299
    102 Atomic Habits                      1002        1      499.00          499
    102 Atomic Habits                      1008        2      499.00          998
    103 Database System Concepts           1003        1      850.00          850
    104 Panchatantra Stories               1004        3      199.00          597
    106 The Kite Runner                    1004        1      350.00          350
    107 Think and Grow Rich                1001        1      275.00          275
    107 Think and Grow Rich                1009        2      275.00          550
    108 Let Us C                           1005        1      450.00          450
    109 Wings of Fire                      1002        2      325.00          650
    110 Harry Potter Stone                 1005        1      599.00          599
    112 Fundamentals of DBMS               1006        1      950.00          950
    113 The Jungle Book                    1007        2      225.00          450
    114 Ponniyin Selvan                    1007        1      699.00          699

15 rows selected.
```

Task 5.4 – Customer-wise Total Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer c
  6  LEFT JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.Customer_Name
  9  ORDER BY c.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                  TOTAL_ORDERS
----------- ------------------------------ ------------
          1 Nivethitha                                1
          2 Hema                                      1
          3 Ravi                                      1
          4 Karthik                                   1
          5 Priya                                     1
          6 Divya                                     1
          7 Arun                                      1
          8 Meena                                     1
          9 Nisha                                     1

9 rows selected.
```

Task 5.5 – Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase_Amount
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.Customer_Name
  9  ORDER BY c.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                  TOTAL_PURCHASE_AMOUNT
----------- ------------------------------ ---------------------
          1 Nivethitha                                      873
          2 Hema                                           1149
          3 Ravi                                            850
          4 Karthik                                         947
          5 Priya                                          1049
          6 Divya                                           950
          7 Arun                                           1149
          8 Meena                                          1297
          9 Nisha                                           550

9 rows selected.
```